

20.2 Electromagnetic Induction

Question Paper

Course	CIE A Level Physics
Section	20. Magnetic Fields
Topic	20.2 Electromagnetic Induction
Difficulty	Medium

Time allowed: 40

Score: /26

Percentage: /100

Question 1a

(i)

Define magnetic flux.

[2]

(ii)

State Faraday's law of electromagnetic induction.

[2]

[4 marks]

Question 1b

A solenoid has a coil C of wire wound tightly about its centre, as shown in Fig. 7.1.

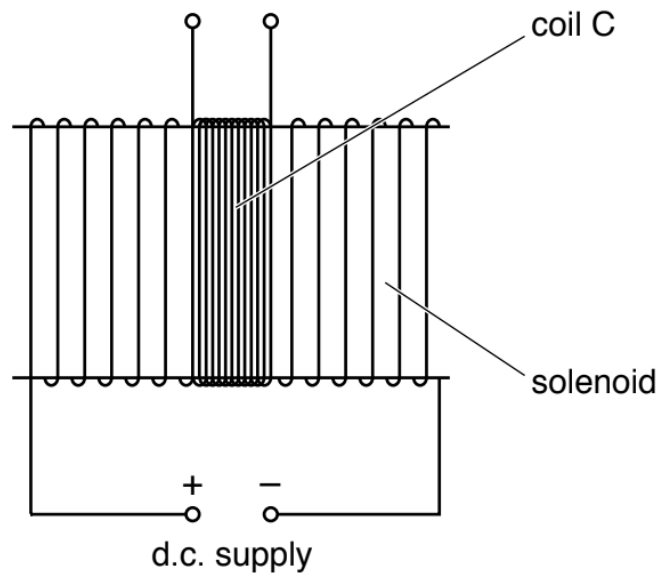


Fig. 7.1

The coil C has 96 turns.

The uniform magnetic flux Φ , in Wb, in the solenoid is given by the expression

$$\Phi = 6.8 \times 10^{-6} \times I$$

where I is the current, in A, in the solenoid.

Calculate the average electromotive force (e.m.f.) induced in coil C when a current of 3.5 A is reversed in the solenoid in a time of 2.4 ms.

e.m.f. = V

[2 marks]

Question 1c

The d.c. supply in Fig. 7.1 is now replaced with a sinusoidal alternating supply.

Describe qualitatively the e.m.f. that is now induced in coil C.

[2 marks]

Question 2a

A small solenoid of area of cross-section $1.2 \times 10^{-3} \text{ m}^2$ is placed inside a larger solenoid of area of cross-section $7.8 \times 10^{-3} \text{ m}^2$, as shown in Fig. 1.1

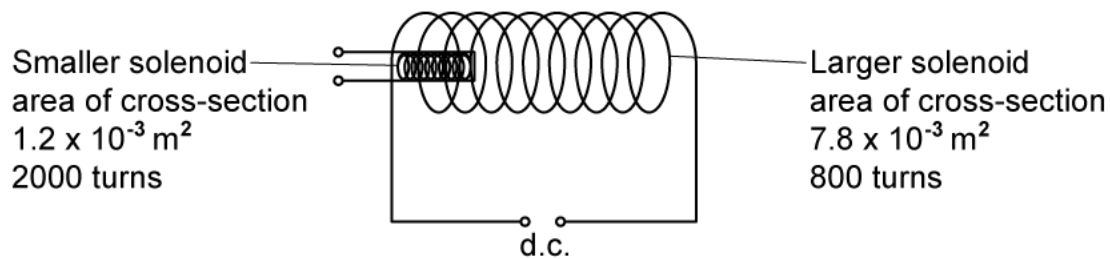


Fig. 1.1

The larger solenoid has 800 turns and is attached to a d.c. power supply to create a magnetic field.

The smaller solenoid has 2000 turns.

Compare the magnetic flux in the two solenoids.

[1 mark]

Question 2b

Compare the magnetic flux linkage in the two solenoids.

[1 mark]

Question 2c

State Lenz's Law of electromagnetic induction.

[2 marks]

Question 2d

The terminals of the smaller solenoid are connected together. The smaller solenoid is then removed from the inside of the larger solenoid.

With reference to magnetic fields, explain why a force is needed to remove the smaller solenoid.

[3 marks]**Question 3a**

Define magnetic flux linkage.

[2 marks]

Question 3b

A square coil of wire of side length 10 cm consists of 5 insulated turns. The coil is stationary in a uniform magnetic field. The plane of the coil is perpendicular to the magnetic field, as shown in Fig. 1.1.

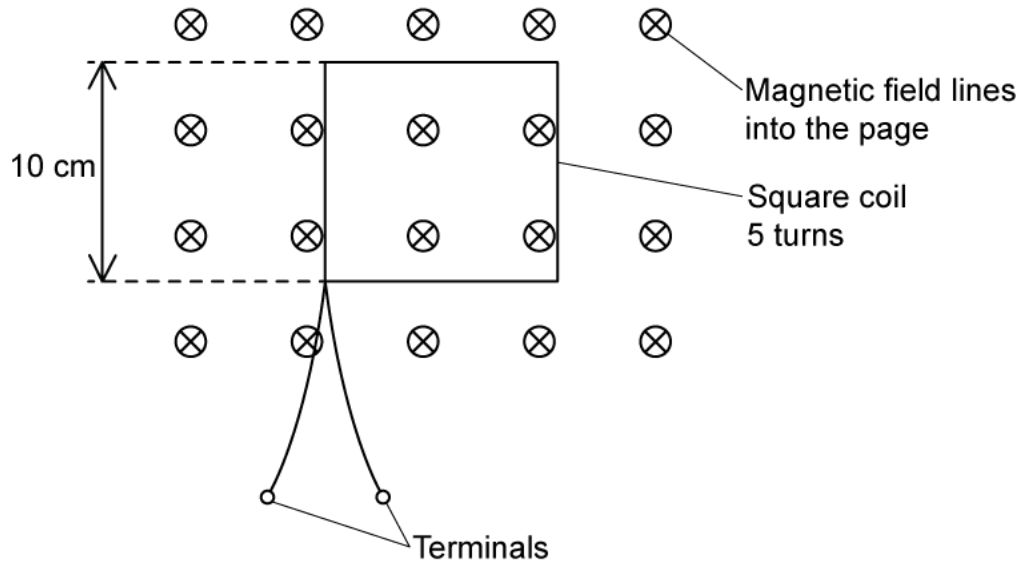


Fig. 1.1

The flux density B of the magnetic field varies with time t as shown in Fig. 1.2.

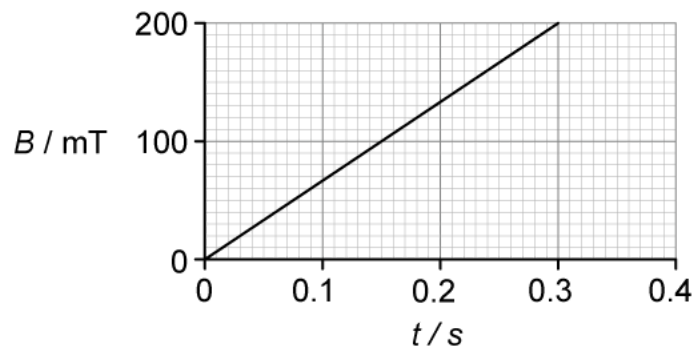


Fig. 1.2

Determine the magnetic flux linkage inside the coil at time $t = 0.30$ s. Give a unit with your answer.

magnetic flux linkage = unit

[3 marks]

Question 3c

(i)

State how the graph in Fig. 1.2 shows that the e.m.f. induced across the terminals between $t = 0$ and $t = 0.30$ s is constant.

[1]

(ii)

Calculate the magnitude of the e.m.f.

e.m.f = V [2]

[3 marks]**Question 3d**

The procedure in (b) is repeated, but this time the terminals of the coil are connected together.

State and explain the effect on the coil of connecting the terminals together during the change of magnetic flux density shown in Fig. 1.2.

[3 marks]

Model Answers: Medium

1a

(a)

(i) Define magnetic flux:

- Magnetic flux is the product of magnetic flux density and cross-sectional area; [1 mark]
- The direction of the magnetic flux is normal to the plane of the cross-sectional area; [1 mark]

OR

- Magnetic flux = magnetic flux density $\times$ area $\times \sin \theta$ [1 mark]
- Where θ is the angle between the direction of the flux and the plane of the area; [1 mark]

(ii) State Faraday's law of electromagnetic induction:

- Faraday's law states that the induced e.m.f is proportional to the rate; [1 mark]
- Of change of magnetic flux linkage; [1 mark]

[Total: 2 marks]

- Use the **revision notes** to learn the definitions

1b

(b) Calculate the average electromotive force (e.m.f.) induced in coil C:

List the known quantities:

- Number of turns in coil C, $N = 96$
- Magnetic flux, $\phi = 6.8 \times 10^{-6} \times I$
- Current in the solenoid, $I = 3.5 \text{ A}$

- Time interval, $t = 2.4 \text{ ms} = 2.4 \times 10^{-3} \text{ s}$

Recall the equation for Faraday's Law:

- $\varepsilon = N \frac{\Delta\phi}{\Delta t}$

Consider the phrase "the current is reversed":

- This means the current is first reduced to zero and then increased to the same value in the opposite direction
- So, the flux linkage must be considered twice

Substitute the known quantities into the equation for Faraday's Law:

- $\varepsilon = 96 \times \frac{(6.8 \times 10^{-6}) \times 3.5 \times 2}{2.4 \times 10^{-3}}$ [1 mark]

Calculate the e.m.f, ε :

- $\varepsilon = 1.904 \text{ V} = 1.9 \text{ V (2s.f.)}$ [1 mark]

[Total: 2 marks]

- Look out for "the **current is reversed**" this tells you to **multiply** flux linkage **by 2**

1c

(c) Describe qualitatively the e.m.f. that is now induced in coil C:

- The e.m.f now induced in coil C is alternating / sinusoidal; [1 mark]
- It has the same frequency as the sinusoidal alternating supply; [1 mark]

[Total: 2 marks]

2a

(a) The magnetic flux is:

- Less in the smaller solenoid; [1 mark]

[Total: 1 mark]

2b

(b) The magnetic flux linkage is:

- Greater in the smaller solenoid; [1 mark]

[Total: 1 mark]

2c

(c) Lenz's Law of electromagnetic induction is:

- The direction of the induced e.m.f.; [1 mark]
- That produces the effects that oppose the change causing it; [1 mark]

[Total: 2 marks]

2d

(d) A force is needed to remove the smaller solenoid because:

- In the smaller solenoid, the change in flux linkage induces an e.m.f. in it; [1 mark]
- The induced current in the smaller solenoid causes a field around it; [1 mark]
- The two fields interact to create an attractive force; [1 mark]

[Total: 3 marks]

3a

(a) The definition of magnetic flux linkage is:

- The product of magnetic flux; [1 mark]
- and the number of turns of a coil; [1 mark]

[Total: 2 marks]

3b

(b) Determine the magnetic flux linkage inside the coil at time $t = 0.30$ s:

List the known quantities:

- Side length of coil, $L = 10 \text{ cm} = 0.1 \text{ m}$
- Number of turns in coil, $N = 5$

Calculate the area of the coil:

- Cross-sectional area, $A = L^2 = 0.1^2 = 0.01 \text{ m}^2$

Use the graph to obtain a value for magnetic flux density, B :

- At $t = 0.3 \text{ s}$, $B = 200 \text{ mT} = 200 \times 10^{-3} \text{ T}$

Use the equation for magnetic flux linkage:

- $\Phi N = BAN$ [1 mark]
- $\Phi N = (200 \times 10^{-3}) \times 0.01 \times 5$ [1 mark]
- $\Phi N = 0.01 \text{ Wb}$ [1 mark]

[Total: 3 marks]

3c

(c)

(i) The graph shows that the e.m.f. induced across the terminals between $t = 0$ and $t = 0.30 \text{ s}$ is constant because:

- It is a straight line / has a constant gradient; [1 mark]

(ii) The magnitude of the e.m.f. is:

List the known quantities:

- Magnetic flux linkage, $N\Phi = 0.01 \text{ Wb}$
- Time, $t = 0.30 \text{ s}$

Use Faraday's Law:

- The induced e.m.f. = rate of change of flux linkage; [1 mark]

- It is a straight line / has a constant gradient; [1 mark]

(ii) The magnitude of the e.m.f. is:

List the known quantities:

- Magnetic flux linkage, $N\Phi = 0.01 \text{ Wb}$
- Time, $t = 0.30 \text{ s}$

Use Faraday's Law:

- The induced e.m.f. = rate of change of flux linkage; [1 mark]
- So, $\text{e.m.f.} = \frac{N\Phi}{t}$
- $\text{e.m.f.} = \frac{0.01}{0.3} = 0.033 \text{ V}$ [1 mark]

[Total: 3 marks]

3d

(d) The effect on the coil of connecting the terminals together during the change of magnetic flux density shown in Fig. 1.2 is:

- The induced e.m.f. causes current flow in the coil; [1 mark]

Any **pair** from:

- Current in the magnetic field causes the forces to act on the coil; [1 mark]
- The opposite sides of the coil are forced inwards; [1 mark]

OR

- Current causes the dissipation of energy in the resistance of the coil; [1 mark]
- The temperature of the coil rises; [1 mark]

[Total: 3 marks]

